

FACULTY
OF
INFORMATION
AND
COMMUNICATION TECHNOLOGY

NATIONAL DIPLOMA: COMPUTER SYSTEMS ENGINEERING

PROJECT DESIGN II: PD2316D

PROJECT PROPOSAL

PROJECT TITLE: **BIOMETRIC STUDENT ATTENDANCE AND VERIFICATION SYSTEM**

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INTRODUCTION

Student attendance is an important part of academic administration, as it provides a record of student participation in lectures, practical sessions, examinations, and other academic activities. In many institutions, attendance is still recorded manually using paper-based registers where students sign their names and receive attendance slips as proof of attendance. Although this method is simple, it poses several challenges, such as inaccurate records, lost/damaged attendance sheets, and time-consuming record management. Retrieving and analyzing historical records can be difficult when dealing with large amounts of paperwork.

This project proposes the design and development of a Biometric student attendance and verification system that will provide a digital alternative to the existing paper-based attendance process. The system will use biometric methods such as fingerprint scanning and facial recognition to verify a student's identity before recording attendance. Once the student has been verified, the system will capture the student's attendance along with relevant information such as timestamps, venue, and session details, creating a digital, traceable attendance record.

The proposed system is intended to improve accuracy, efficiency, and reliability of attendance [1], [2]. Management. By using biometric verification, the system ensures that attendance is recorded for the correct student and reduces the possibility of fraudulent records. Digital storage will make the information easier to manage, retrieve, and analyze while saving costs on buying paper for attendance sheets.

The project will combine biometric sensing, embedded hardware, software, and database technologies to develop a complete solution. By addressing the limitations of the current manual process, the proposed system aims to provide a more secure, efficient, and accountable method of recording and managing students' attendance.

PROJECT OBJECTIVES

The main objective of the project is to design and develop a system that provides a reliable digital alternative to the current paper-based examination attendance process.

Specific objectives

- Develop a system that can verify students using biometric methods such as fingerprint scanning and facial recognition.
- Automatically record students' attendance after successful verification.
- Capture the exact date-time and venue of each entry to create a traceable record.
- Reduce the possibility of proxy attendance and duplicate attendance entries.
- Provide authorized users with access to records for viewing, searching, and monitoring.
- Allow attendance information to be retrieved and used to generate reports when required.
- Integrate biometric hardware, software, and database technologies into one functional system.
- Test the system for accuracy, reliability, and usability.

The project will be implemented in stages, beginning with system analysis and design, followed by the development of biometric verification components, databases and software integration, testing, and final evaluation. A working prototype will be developed and demonstrated.

PROJECT DESCRIPTION

The project is a Biometric student attendance and verification system designed to provide a more efficient way of recording and managing student attendance. The system is an original concept intended for use in academic environments such as universities, colleges, training centers, and other institutions where attendance and access control must be monitored.

The target users of the system will mainly include students, lecturers, administrators, and other authorized staff. Students will use the system to mark their attendance using biometric verification, while authorized staff will be able to access and review attendance information when required.

The system will cover the registration of student information, biometric enrolment, student verification, automatic attendance recording, date and time stamping, storage of records, and retrieval of said records. It will provide basic reporting functionality to allow authorized users to review information for a specific student, class, session, or exam.

The scope of the project will focus on attendance management and identity verification. It will not replace existing institutional systems such as student registration but will add to it by using current student data to help with identity validation and the acquisition of new student data.

The main features of the system will therefore include biometric student verification, automatic attendance capture, dated records, centralized digital storage, and basic reporting. The system will be designed in a way that could allow expansion or integration with existing institutional systems.

PROJECT TECHNOLOGY

Software's

1. Python – main programming language for biometric processing.
2. OpenCV – used for capturing, detecting, and recognizing faces from the camera [3].
3. FastAPI – backend framework; it will create the API that connects the biometric system, database, and user interface [7].
4. PostgreSQL – Database for storing student details, attendance sessions, and timestamps [6].
5. Programming languages (HTML, CSS, NEXTJS) – used to develop the user interface.
6. Arduino IDE – Used to program the microcontroller that will interface with the biometric hardware such as fingerprint scanners and face recognition cameras.
7. Adafruit fingerprint sensor library – handles communication with compatible fingerprint modules.
8. VS Code IDE – used to code the backend.
9. KICAD – used to design the PCB and packaging.

Hardware

1. ESP32 development board – main embedded controller. It will communicate with and control all the external sensors and send feedback to the backend [4].
2. Fingerprint module – captures and matches fingerprints for student verification [5].
3. USB camera module – captures student faces for facial recognition.
4. Laptop/desktop – runs the backend, database, and administrative dashboard.
5. Power supply – supplies power to the esp32 and connected components.
6. LED indicator – provides simple feedback such as successful verification or verification failure.
7. Buzzer – provides an audible indication when attendance is captured successfully or fails.
8. Breadboard – used for prototyping.
9. LCD Display – shows messages or instructions such as place fingers, or verification status.
10. Jumper wires – used to connect the different modules during prototyping.

TIMETABLE

Activity	Description of work	Start and end dates
Project proposal	Define the project problem, objectives, scope, tech stack, budget and overall implementation plan.	01-Aug-2026 – 20-Aug-2026
System research and requirement analysis	Researching biometric systems, identifying functional and non- functional requirements, suitable hardware selection, define system requirements.	21-Aug-2026 – 26-Aug-2026
System design	Develop the system architecture, database design, UI design, hardware layout, data flow and validation process	27-Aug-2026 – 08-Sep-2026
Prototype hardware development	Configure the esp32, fingerprint sensor, camera, display, indicators and other hardware requirements.	09-Sep-2026 – 18-Sep-2026
Software and database	Develop the backend, database, attendance, recording, validation, student registration, admin functions	09-Sep-2026 – 25-Sep-2026
System integration testing	Biometric verification, timestamp recording, verification, database storage, duplicate prevention and system reliability must be tested as a unit.	26-Sep-2026 – 30-Sep_2026
User documentation	Prepare the user manual explaining system setup and operation	01-Oct-2026 – 08-Oct-2026
System refinement	Fault finding during testing and optimizing based on user feedback	09-Oct-2026 – 16-Oct-2026
Presentation preparation	Preparing demonstration material, slides, diagrams, and the project demonstration.	17-Oct-2026 – 19-Oct-2026
Mock presentation	Present the working project for preliminary assessment and obtain feedback for improvements.	20-Oct-2026 – 21-Oct-2026
Technical report finalization	Completing the technical report, including design, implementation, testing, results, challenges, and conclusions.	21-Oct-2026 – 25-Oct-2026
Final improvements and testing	Implement feedback from the mock presentation and perform final system testing.	26-Oct-2026 – 31-Oct-2026
Final presentation preparation	Finalize the presentation setup and demo setup.	01-Nov-2026 – 02-Nov-2026
Final project presentation	Present and demonstrate the complete Biometric student attendance Verification system.	03-Nov-2026 – 04-Nov-2026

BUDGET

State the proposed costs and budget of the project. Also include information on how you intend to manage the budget.

Items	Supplier	Quantity	Unit price	Total
ESP32 development board	Communica	1	R230.00	R230.00
Fingerprint module	Takealot	1	R470	R470
USB camera module	Takealot	1	R399.92	R399.92
Power supply	Takealot	1	R300.00	R300.00
LED indicator	Communica	3	R2.95	R8.85
Buzzer	Communica	1	R39.50	R39.50
Development board	Communica	1	R47.83	R47.83

Jumper wires	Communica	2	R20.87	R41.74
LCD Display	Communica	1	R86.96	R86.96
Double sided PCB kit	Takealot	1	R388.00	R388.00
				Total = R2012,8

CONCLUSION

The project aims to provide a practical digital alternative to the current paper-based attendance and access control process of institutions. By combining fingerprint and facial recognition with digital record-keeping, the system will improve the accuracy, traceability, and efficiency of attendance management.

The project will demonstrate the integration of biometric hardware, embedded systems, software deployment, networking, and database technologies into a single working solution. The system will allow student attendance to be captured automatically with a corresponding date and time, while giving authorized users access to the records when required.

The outcome of this will be a functional device that demonstrates how biometric technology can be applied to student attendance management and the potential for access control.

REFERENCE

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[7]. FastAPI (2023). FastAPI. [online] fastapi.tiangolo.com. Available at: https://fastapi.tiangolo.com/ [Accessed 19 Aug. 2026].

[8]. vitejs. (2025). Getting Started. [online] Available at: https://vite.dev/guide/. [Accessed 19 Aug. 2026].